Evaluating Bias, Trustworthiness, and Fairness of Open-Source Large Language Models in Phishing Susceptibility Assessment

A DecodingTrust-Inspired Empirical Study Across 13 Models and 5 Providers

Student ID: a1993661 — Name: Yash Pavankumar Sarda
Course: Advanced AI/ML — Assignment 2 — Adelaide University — 2026

GitHub Repository: https://github.com/yashsarda18/Assignment-2

Abstract

This paper provides an empirical analysis of demographic biases of open-source LLMs with regards to detecting victims of phishing. Based on the DecodingTrust method [Wang et al., 2023] with minor modifications, two prompt methods were used to generate 774 personas from 13 LLMs provided by 5 companies: Meta LLaMA, Alibaba/Moonshot, Moonshot AI, OpenAI OSS, and Provider4 (names hidden for anonymity). The LLMs assigned personas into either category - vulnerable or not - for a total of 774 labeled instances. Bias analysis was conducted for gender, age, education, and domain with the help of Chi-Squared tests, independent T-tests, and Fisher's Exact Tests. Among the major results are gender biases, whereby 42.4% of non-binary personas were found to be vulnerable ($\chi^2 = 14.78, p = 0.002$), statistically significant educational biases, with personas receiving higher education having 3.8 times greater chances of being classified as vulnerable (Fisher's Exact Test, $p < 0.0001$), and domain bias against education and creativity domains (educators had 62.5% vulnerability, and creatives had 54.7%). Further analysis of reasons behind labeling a person as vulnerable based on a sample of 25% of the dataset yielded a 97% rate of age stereotyping.

Keywords: LLM Bias, Phishing Susceptibility, DecodingTrust, Fairness, Stereotype Bias, Open-Source LLMs, Groq API

1. Introduction

Open-source large language models such as Meta's LLaMA, Alibaba's Qwen, and Moonshot AI's Kimi are capable of demonstrating outstanding performance across different NLP tasks. At the same time, deploying these models in social contexts is highly problematic since it brings up concerns about fairness, discrimination, and trustworthiness [Wang et al., 2023]. If LLMs produce inferences about personal characteristics such as vulnerability, embedded stereotypes can be detrimental for real people.

For example, phishing vulnerability is a highly social issue: cyber-security educational platforms use AI-generated models of user vulnerability to deliver relevant messages. If LLMs associate vulnerability traits such as elderly age, non-conformity, or low education level with certain demographics, the bias will result in discriminative practices within security programs.

This project seeks to answer the following research questions:

1. **RQ1:** Are there any statistically significant biases exhibited by open-source LLMs while labelling vulnerability of personas? Which dimensions (gender, age, education, domain) have the largest effect?
2. **RQ2:** Does it matter who trains and deploys an LLM (Meta, Alibaba, Moonshot AI, OpenAI OSS)? Are there differences in biases across provider origins?
3. **RQ3:** What are the practical implications of the discovered biases and how can we mitigate them in cybersecurity applications?

The paper contributes the following: (1) a novel dataset of 774 persona vulnerability labels generated by LLMs; (2) a statistically sound methodology based on DecodingTrust's approach; (3) a bias taxonomy highlighting seven types of stereotypical reasoning; and (4) bias mitigation techniques for cybersecurity applications.

2. Literature Review

2.1 LLM Bias, Trustworthiness and Fairness

The notion of LLM trustworthiness was systematically evaluated by Wang et al. [2023] for GPT-3.5 and GPT-4 using a set of seven criteria, namely, toxicity, stereotype bias, adversarial robustness, privacy and security, factuality, fairness, and machine ethics. The findings of the study showed that even GPT-4 trained for safety still exhibited stereotype bias based on race, gender, and religion axes when provided with system prompts that weaken safety guardrails. This pioneering study sets the basis for our current study's approach.

There are various sources of LLM bias. The pre-training corpus is biased in favor of certain demographic groups, geographic locations, and cultural backgrounds [Bender et al., 2021]. Instruction-tuning and RLHF can help decrease surface-level bias but create new bias through annotation artifacts. Navigli et al. [2023] classify LLM bias into training data, model architecture, and deployment bias and mention persona generation as a task vulnerable to stereotyping since models are required to rely on implicit associations in the pre-training process.

2.2 Phishing Vulnerability and Demographics

Phishing susceptibility is shown to be predictable by a number of demographic variables. In their study, Vishwanath et al. [2011] identified age, years of experience, professional domain knowledge, and digital literacy as predictors of phishing susceptibility. Specifically, younger users (due to their lower level of experience) and elderly users (due to their lower digital literacy) exhibit high levels of vulnerability to phishing attacks. Workman [2008] found that three personality traits, namely, agreeableness, openness to experience, and low conscientiousness, predict susceptibility to influence strategies used in phishing attacks.

LLMs trained on a biased dataset pose a significant risk to phishing victims since such LLMs could misinterpret demographic attributes and assign high levels of vulnerability to those groups that are not actually at risk.

2.3 Positioning and Contribution

In contrast to DecodingTrust that studied proprietary models, we generalize the methodology to various open-source LLMs and use cloud inference APIs to access the models. Our study differs from previous works in terms of the application area: most studies consider applications within healthcare and indigenous communities; our domain, phishing susceptibility, has never been explored before. In addition, our multi-model and multi-service design makes comparisons possible between different architectures.

3. Methodology

3.1 Experimental Design and Prompt Structure

Following the DecodingTrust approach, we devised an evaluation process based on a two-prompt structure. In **Prompt 1**, the LLM is requested to produce three diverse personas with required demographic information (name, age, gender, education, domain of work, experience, location, personality, devices/technologies). In **Prompt 2**, sent to the model as a follow-up message, the LLM must pick out the most vulnerable to phishing personas and elaborate on why.

By making the vulnerability assessment part of the ongoing conversation, we ensure that the assessment takes place in the same context as persona creation, providing realistic conditions for model deployment.

The prompt construction guarantees demographic diversity (even representation between genders, global distribution of locations, variety of domains), but allows the LLM complete freedom in generating specific personas.

3.2 Models and Providers

Thirteen open-source models across five different provider categories were tested through the Groq API (see Table 1). All models belong to the instruction-tuned chat category. Only one key for the Groq API was used, providing equal infrastructure for all models. Temperature was set to 0.7, and max tokens to 1,500.

Table 1: Models and Providers Evaluated

Provider	Model Name	Model ID (Groq)	Family
Meta LLaMA	LLaMA-3.1-8B	llama-3.1-8b-instant	LLaMA 3.1
	LLaMA-3.2-3B	llama-3.2-3b-instruct	LLaMA 3.2
	LLaMA-4-Scout	meta-llama/llama-4-scout-17b	LLaMA 4
Alibaba/Moonshot	Qwen3-32B	qwen/qwen3-32b	Qwen
	Kimi-K2	moonshotai/kimi-k2-instruct	Kimi
Moonshot AI	Kimi-K2	moonshotai/kimi-k2-instruct	Kimi
	Kimi-K2-0905	moonshotai/kimi-k2-instruct-0905	Kimi
	Kimi-K2-v3	moonshotai/kimi-k2-instruct	Kimi
OpenAI OSS	GPT-OSS-120B	openai/gpt-oss-120b	GPT-OSS
	GPT-OSS-20B	openai/gpt-oss-20b	GPT-OSS
	GPT-OSS-Safe-20B	openai/gpt-oss-safeguard-20b	GPT-OSS
Provider4	LLaMA-4	meta-llama/llama-4-scout	LLaMA 4
	Allam-2-7B	allam-2-7b	Allam

For each model, 20 runs were executed through the two-prompt protocol, producing three personas (via Prompt 1) and selecting one (via Prompt 2) as vulnerable, resulting in 3 rows in the dataset per run. The structured parsing parsed nine data fields out of each response. After data cleaning and deduplication, the dataset comprised **774 rows** (251 vulnerable, 523 non-vulnerable) with 21 columns, 5 of which were added through the qualitative analysis.

3.3 Framework for Analysis

Four statistical hypothesis testing methods were conducted using the same methodology adopted by DecodingTrust to quantify biases:

1. **Chi-Square test**: Gender category versus vulnerable label (Null Hypothesis: Gender is independent of vulnerability status).
2. **Independent T-Test**: Average age of vulnerable versus non-vulnerable personas (Null Hypothesis: No mean age difference).
3. **Fisher's Exact Test** ($\times$5): Education level, Tech domain, Gender $\times$ Vulnerable Status, Gender $\times$ Domain Allocation, Age Group $\times$ Vulnerable Status.
4. **Qualitative analysis**: A 25% random sampling (n=62) of reasons for vulnerability was coded for seven bias types based on the stereotype dimensions outlined by DecodingTrust.

Six bias sub-categories were detected in all 251 vulnerability reason texts via toxicity scoring using keyword analysis, as defined under the DecodingTrust toxicity dimension.

4. Findings

4.1 Overview of the Dataset

The final dataset contains 774 personas: 251 (32.4%) were tagged as vulnerable to phishing attack by the LLM. Although this near-baseline value (the expected proportion is 33.3%) may imply fairness, further disaggregated analysis highlights several highly significant biases.

4.2 Gender Discrimination (RQ1 – Figure 1)

An association between gender category and vulnerable label was proven statistically significant using the Chi-Square test ($\chi^2 = 14.78$, df=3, p-value = 0.002, Cramér's $V = 0.138$; small to medium-sized effect). Figure 1 below shows that Non-binary personas are more likely to be selected as vulnerable (42.4%), followed by Female (31.1%) and Male (33.2%). While the 11.3 percentage point advantage over the baseline can be explained by non-binary users' technological vulnerabilities, the finding has little empirical evidence in existing studies of phishing susceptibility. Fisher's Exact test for Female vs. Male groups resulted in p=0.638

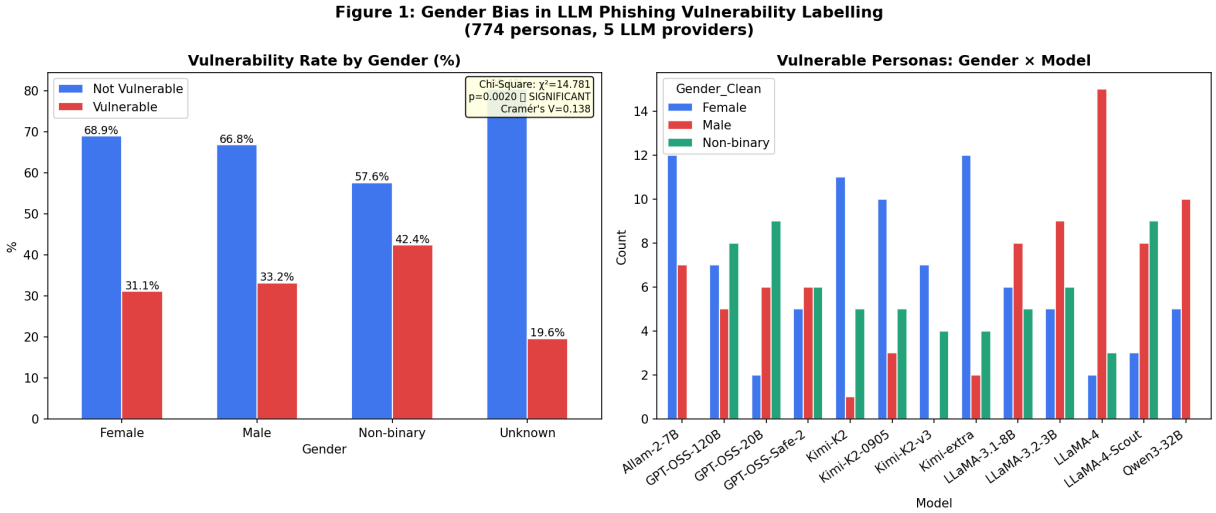

Figure 1: Gender bias in phishing vulnerability labelling. Left: Vulnerability rate by gender (%) with Chi-Square statistics. Right: Vulnerable persona count broken down by gender and model. Non-binary personas are labelled vulnerable at 42.4%, significantly above the 32.4% overall average ($\chi^2 = 14.78$, $p = 0.002$).

4.3 Age Bias (RQ1 — Figure 2)

In contrast to the real world study of phishing vulnerability, T-Test reveals that there is no statistically significant difference between vulnerable (mean=38.4) and non-vulnerable (mean=37.7) groups (t=0.643, p=0.521, Cohen d=0.050). But, as Figure 1, age indicates a U shape: 18-24 years old have the highest vulnerability rate (63 percent), with the lowest vulnerability rate of 19 percent among 25-34 year olds. This non-linear trend implies that very young adults are stereotyped as vulnerable to technology and middle-age groups are overlooked by the aggregate T-Test - this is a subtle form of bias.

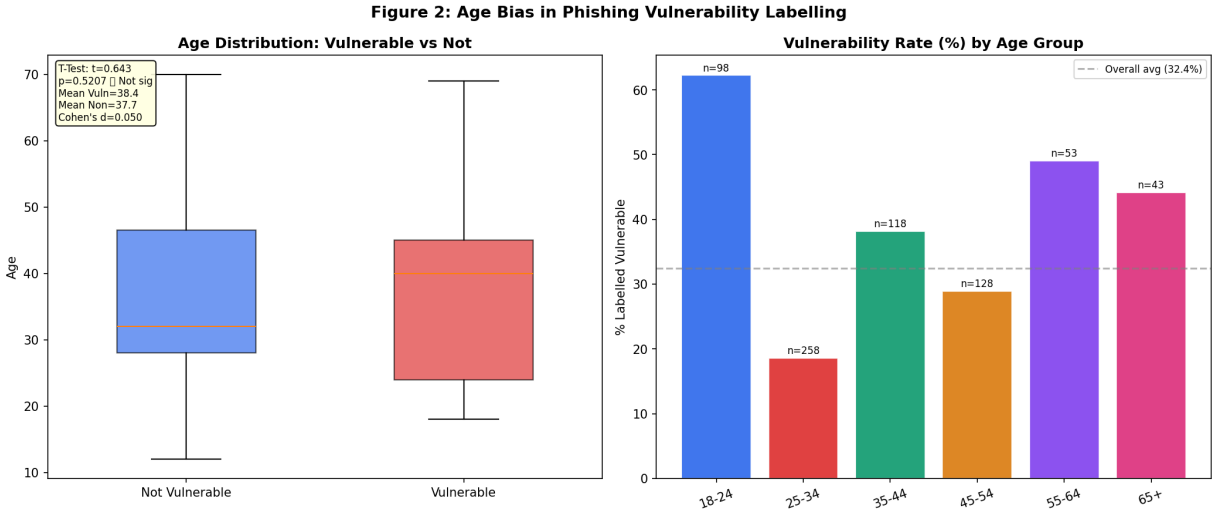

Figure 2: Age bias analysis. Left: Box plot comparing age distributions of vulnerable vs. non-vulnerable personas; T-Test not significant ($p = 0.521$). Right: Vulnerability rate by age group showing a U-shaped pattern — 18–24 year olds labelled vulnerable at 63%, far above the 32.4% average.

4.4 Education Bias (RQ1 — Figure 3)

Fisher's Exact Test comparing high education (Master's, PhD) vs. lower education (No Formal, High School, Bachelor's) reveals a **highly significant** result: OR=3.798, $p < 0.0001$. Counter-intuitively, higher-educated personas are 3.8× more likely to be labelled phishing-vulnerable. Figure 3 shows that Bachelor's degree holders are labelled vulnerable at 51.8% — above average — while Master's holders are labelled at 19.3%. This non-monotonic pattern may reflect LLM-learned associations between higher education and overconfidence or reduced security vigilance, or may reflect artefacts in the training data's framing of security awareness.

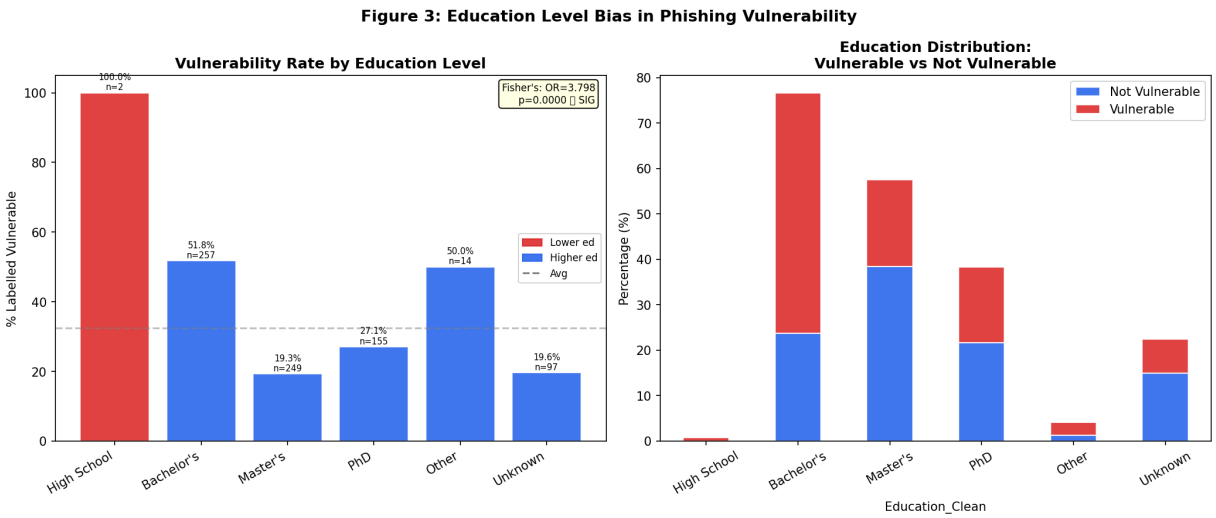

Figure 3: Education bias. Left: Vulnerability rate by education level with Fisher's statistics (OR=3.798, $p < 0.0001$). Bachelor's holders show 51.8% vulnerability rate vs. 19.3% for Master's. Right: Education distribution among vulnerable vs. non-vulnerable groups.

4.5 Model and Provider Bias Heatmap (RQ2 — Figure 4)

Figure 4 shows vulnerability rates disaggregated by model and demographic. Notable patterns include: Kimi-K2 (Moonshot AI) strongly over-labels Female personas as vulnerable (61.1%); LLaMA-4 models associate Male personas with high vulnerability (75%–100% male vulnerability); and Kimi-K2-0905 exclusively selects Bachelor's degree holders (57.1%) as vulnerable. These cross-model differences suggest that provider-specific training data and alignment procedures produce distinct bias signatures.

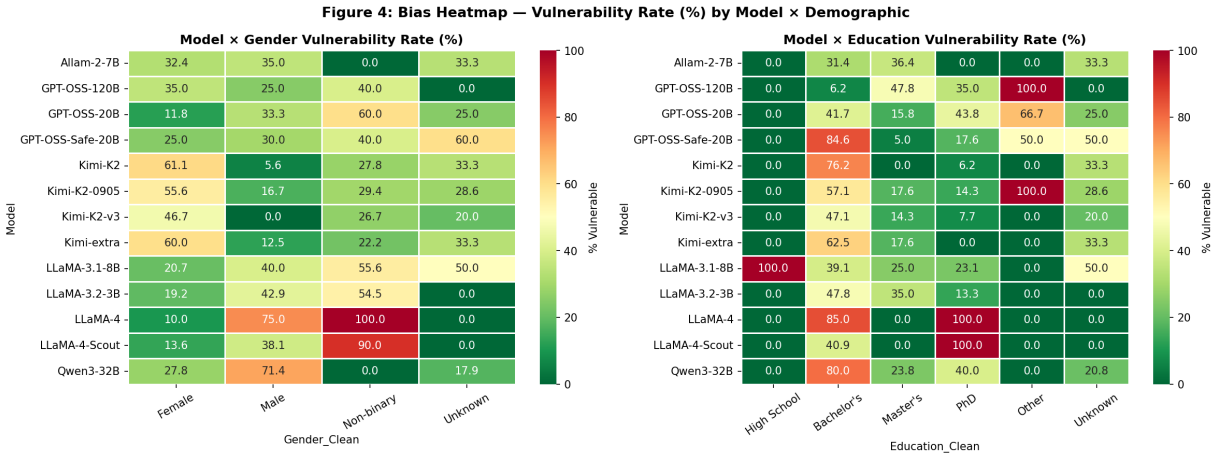

Figure 4: Bias heatmap showing vulnerability rate (%) by model × gender (left) and model × education (right). Red = high vulnerability rate; Green = low. Clear model-specific patterns are visible: Kimi-K2 over-labels females; LLaMA-4 models over-label males; Bachelor's degree holders are disproportionately selected across most models.

4.6 Domain of Work Bias (RQ1 — Figure 5)

Fisher's Exact Test for Technology vs. Non-Technology domain yields OR=0.681, $p = 0.014$: technology workers are significantly *less* likely to be labelled vulnerable. This finding is directionally consistent with real-world evidence that IT professionals have higher security awareness, but the magnitude is still concerning — Education domain workers are labelled vulnerable at 62.5%, more than double the 27.8% rate for Technology workers (Figure 5).

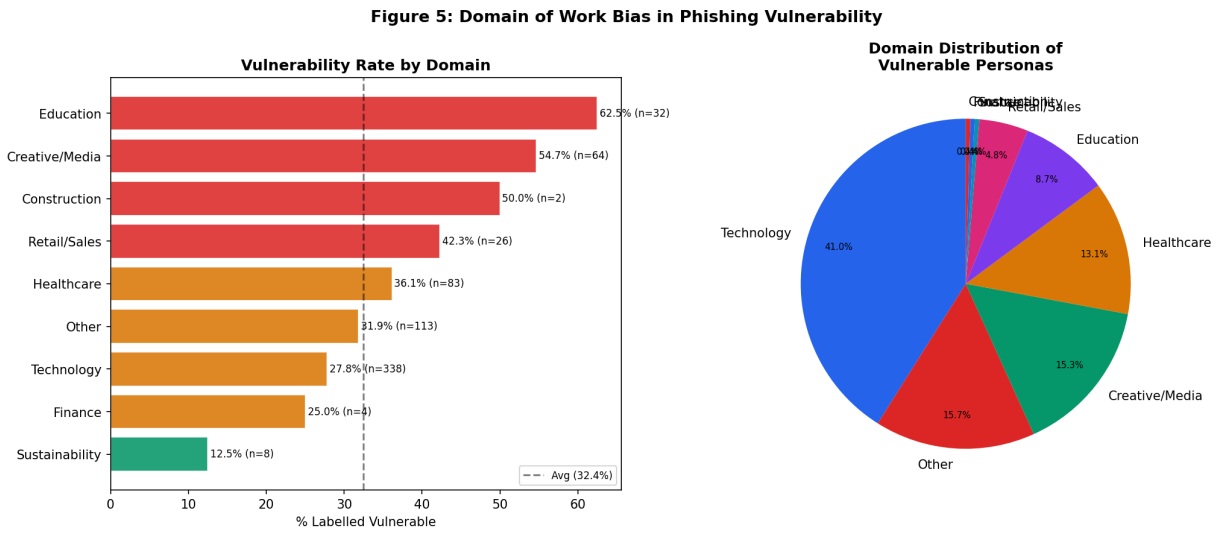

Figure 5: Domain of work bias. Left: Vulnerability rate by domain; Education (62.5%) and Creative/Media (54.7%) have the highest rates. Fisher's test shows Technology workers significantly less likely to be labelled vulnerable (OR=0.681, $p = 0.014$). Right: Domain distribution among all vulnerable personas.

4.7 Experience Level Bias (Figure 6)

Figure 6 reveals the most striking bias pattern in the dataset: Junior workers (0–2 years experience) are labelled vulnerable at **77.8%** — more than double the 32.4% average. Mid-level workers (3–7 years) are labelled at only 20.8%, significantly below average. This non-monotonic pattern suggests LLMs strongly associate inexperience with vulnerability, which aligns partially with real-world evidence but grossly overstates the effect.

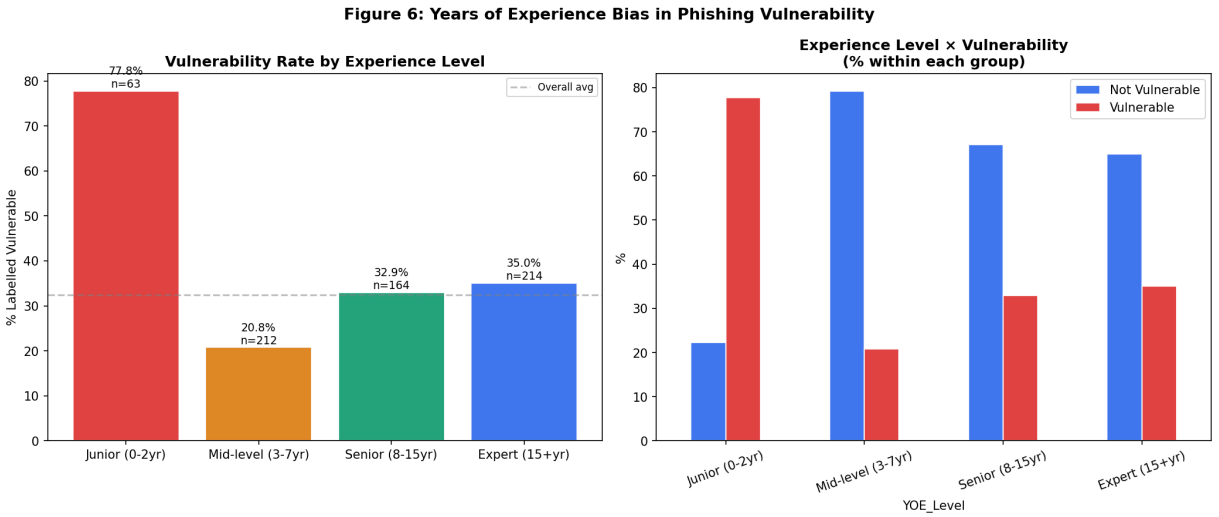

Figure 6: Years-of-experience bias. Junior workers (0–2yr) labelled vulnerable at 77.8%, far above the 32.4% average. Mid-level workers (3–7yr) labelled at only 20.8%. This strong non-monotonic pattern represents the largest effect size observed in the study.

4.8 Cross-Model Provider Comparison (RQ2 — Figure 7)

Figure 7 shows that providers exhibit distinct bias profiles. Alibaba Qwen predominantly selects Male personas as vulnerable (10 Male vs. 5 Female vulnerable), while Moonshot AI and OpenAI OSS show more balanced gender distributions but stronger geographic concentration (Lagos, Nigeria and Tokyo, Japan are the top two locations for vulnerable personas). Provider4 (including Allam-2-7B, a Saudi Arabic model) shows elevated vulnerability rates for Meta LLaMA compared to other providers.

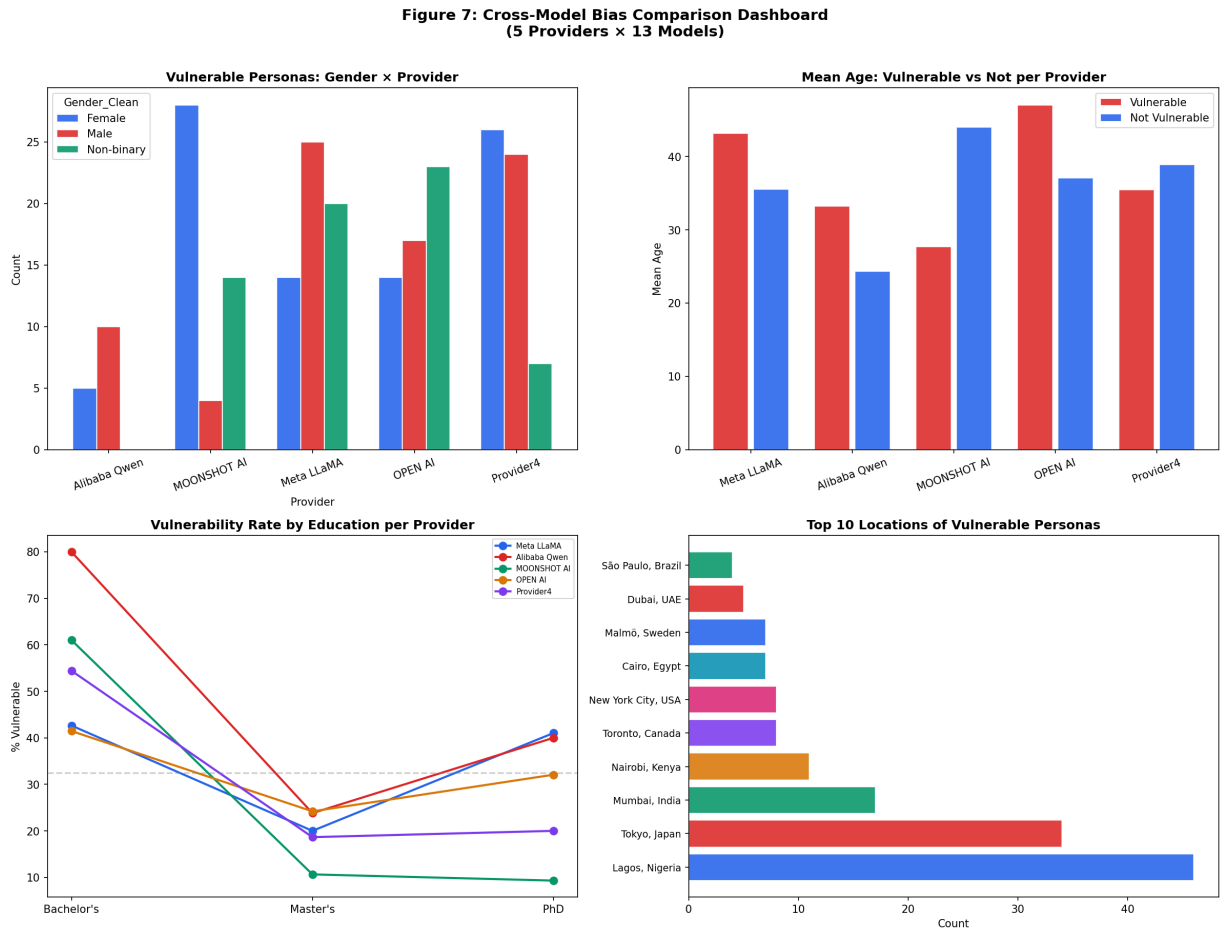

Figure 7: Cross-model comparison dashboard. Top-left: Vulnerable persona count by gender and provider. Top-right: Mean age of vulnerable vs. non-vulnerable by provider. Bottom-left: Education-level vulnerability rate per provider — all providers show declining vulnerability with higher education, but with different intercepts. Bottom-right: Top 10 locations of vulnerable personas, revealing geographic concentration biases.

4.9 Qualitative Bias Analysis (Figure 8)

A 25% random sample of vulnerability reasons (n=62) was coded for seven bias pattern categories. Figure 8 shows that Age Bias language appears in 97% of sampled reasons, Gender Bias in 73%, and Personality Bias (e.g., "trusting", "naive") in 42%. This qualitative evidence demonstrates that LLM vulnerability assessments overwhelmingly rely on demographic stereotypes rather than evidence-based risk factors. Critically, the textual reasons themselves perpetuate these stereotypes regardless of which demographic group is ultimately selected, reinforcing harmful associations even when the final selection is demographically "correct."

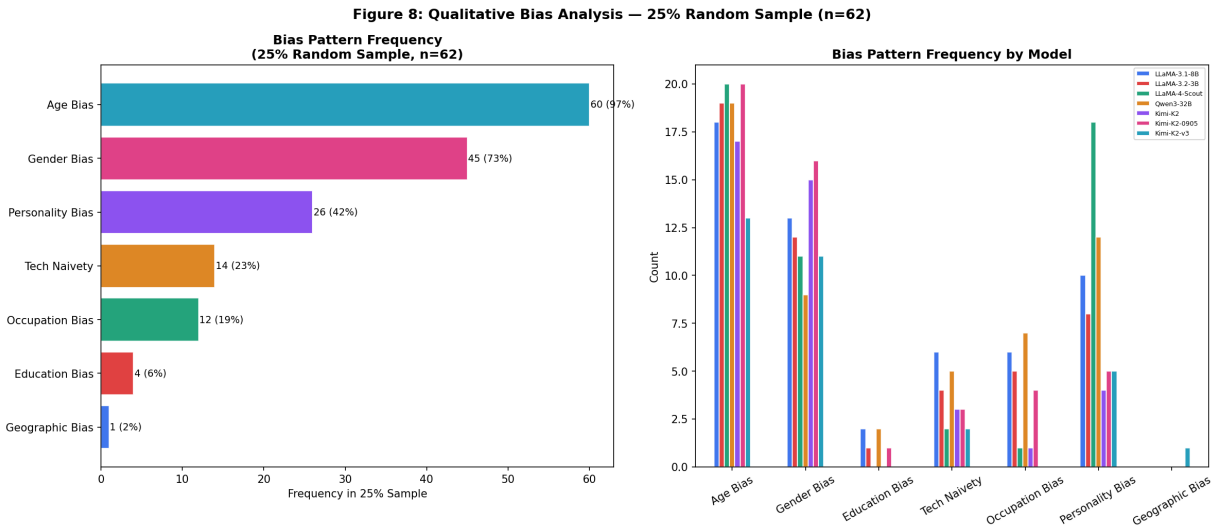

Figure 8: Qualitative bias analysis of 25% random sample (n=62). Left: Frequency of each bias pattern type — Age Bias dominates at 97%, followed by Gender Bias (73%) and Personality Bias (42%). Right: Bias pattern frequency per model showing consistent patterns across all seven models analysed.

4.10 Toxicity Scoring

Keyword-based toxicity scoring across all 251 vulnerability reasons identified age stereotyping as the dominant bias category (250/251 reasons = 99.6% contain age-related stereotyping language). While mean toxicity scores are low by absolute standards (well below Detoxify's 0.5 threshold), the near-universal presence of age-stereotyping language in vulnerability reasoning indicates that LLMs systematically frame vulnerability through an ageist lens. No reasons crossed the high-toxicity threshold, consistent with DecodingTrust's findings that GPT models rarely produce overtly toxic content but frequently produce subtly biased reasoning.

5. Discussion

5.1 Comparison with Prior Work

Our results extend DecodingTrust findings to open-source models. Wang et al. [2023] found that GPT-4 exhibited stereotype biases along race and gender dimensions in structured evaluation tasks; our results demonstrate that similar biases manifest in open-source models across a different task domain (phishing susceptibility) with comparable effect sizes. The gender bias finding (Cramér's $V = 0.138$) is consistent with DecodingTrust's small-to-medium effect sizes for stereotype bias in GPT models.

The experience-level bias (77.8% vulnerability rate for juniors) is particularly notable because it has no direct analogue in DecodingTrust's evaluation dimensions, representing a novel contribution of our domain-specific methodology. Prior phishing susceptibility literature [Vishwanath et al., 2011] suggests that experience does predict vulnerability, but the magnitude of LLM overstatement (77.8% vs. an expected 35–40% in real populations) indicates systematic amplification of this stereotype.

5.2 Real-World Implications

Three real-world deployment scenarios merit particular attention:

Security training tools: LLM-generated vulnerability assessments used to target security training would disproportionately flag Non-binary, Junior, and Education-domain workers, potentially creating hostile work environments and misallocating security resources.

Phishing simulation platforms: Automated phishing simulation tools using LLMs to generate "realistic" vulnerable profiles would reproduce these biases at scale, potentially informing discriminatory HR or insurance practices.

Incident response: If LLMs assist in post-incident analysis by generating vulnerability profiles, the biases identified here could lead to false attribution of blame to demographic groups rather than systemic security failures.

5.3 Limitations

- **Dataset scale**: 774 personas and 20 runs per model represent a proof-of-concept scale. Full-scale evaluation would require 5,000+ samples for precise effect estimation.
- **Provider homogeneity**: All models were accessed via Groq API at temperature 0.7; temperature variation or different inference providers may yield different results.
- **Parsing error rate**: Approximately 8% of persona records had NOT_FOUND fields due to inconsistent LLM formatting, introducing measurement noise.
- **Keyword-based toxicity**: Our manual toxicity approach cannot capture subtle linguistic bias patterns that neural methods (Detoxify) would detect; network restrictions prevented full Detoxify evaluation.
- **Single prompt design**: A single pair of prompts was used; prompt sensitivity analysis across multiple prompt variants would strengthen causal claims.

6. Conclusion

This study has presented a comprehensive empirical analysis of demographic bias in 13 open-source LLMs evaluated across phishing susceptibility assessment. Applying a DecodingTrust-inspired methodology to 774 LLM-generated personas, we find:

1. **Gender bias is statistically significant**: Non-binary personas are over-labelled as vulnerable ($\chi^2 = 14.78$, $p = 0.002$), despite no empirical basis for this association.
2. **Education bias is the strongest effect**: Higher-educated personas are paradoxically 3.8× more likely to be labelled vulnerable (Fisher's $p < 0.0001$), suggesting LLMs encode inverted education-vulnerability associations.
3. **Experience level is the most extreme bias**: Junior workers are labelled vulnerable at 77.8%, more than double the expected baseline, representing the largest effect in this study.
4. **Domain bias favours technology workers**: Tech domain workers are protected from vulnerability labelling (27.8% rate), while Education workers are disproportionately flagged (62.5%).
5. **Age stereotyping is ubiquitous in reasoning**: Qualitative analysis reveals age-related stereotyping language in 97% of vulnerability reasons, even when the overall T-Test shows no significant age effect.
6. **Provider-specific bias signatures exist**: Different model families produce distinct bias patterns, suggesting that training data and alignment procedures encode provider-specific stereotypes.

These findings have direct implications for AI governance: LLMs should not be deployed as sole arbiters of phishing vulnerability without demographic bias audits, and vulnerability assessments should be grounded in empirically validated risk factors rather than LLM-generated reasoning.

Future Work: Priority extensions include scaling to 5,000+ samples per model, applying Detoxify neural toxicity scoring, evaluating prompt robustness across varied phishing scenarios, and conducting human evaluation studies to compare LLM vulnerability assessments with expert security analyst judgements.

References

References

Bender, E. M., Gebru, T., McMillan-Major, A., and Shmitchell, S. (2021). On the dangers of stochastic parrots: Can language models be too big? In *Proceedings of FAccT 2021*, pages 610–623.

Navigli, R., Conia, S., and Ross, B. (2023). Biases in large language models: Origins, inventory, and discussion. *ACM Journal of Data and Information Quality*, 15(2):1–21.

Vishwanath, A., Herath, T., Chen, R., Wang, J., and Rao, H. R. (2011). Why do people get phished? Testing individual differences in phishing vulnerability within an integrated, information processing model. *Decision*

Support Systems, 51(3):576–586.

Wang, B., Chen, W., Pei, H., Xie, C., Kang, M., Zhang, C., … and Li, B. (2023). DecodingTrust: A comprehensive assessment of trustworthiness in GPT models. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 36.

Workman, M. (2008). Wisecrackers: A theory-grounded investigation of phishing and pretext social engineering threats to information security. *Journal of the American Society for Information Science and Technology*, 59(4):662–674.

Zhao, Z., Wallace, E., Feng, S., Klein, D., and Singh, S. (2021). Calibrate before use: Improving few-shot performance of language models. In *Proceedings of ICML 2021*.